

Task 1

a) What will be the **change of the optimal solution** if we **produce 100 pcs of ponies**?

b) What will be the **change of the total profit**?

1. Identify which variable represents **ponies**: ponies (x2)
2. In the final table find column related to ponies:

1,5
0
-0,5
2

3. Divide **change** of 100 pcs by 1000 pcs (toy's unit of measure) = 0,1
4. Copy **final RHS**:

0,7
0,2
0,1
5

5. Proceed calculation (there is negative (-) sign due to the fact that once ponies are produced, it will negatively influence optimal solution):

0,7
0,2
0,1
5

-0,1

1,5
0
-0,5
2

=

New final RHS
and profit

0,55
0,2
0,15
4,8

c) What is the **maximal number of ponies** that can be produced **without changing the optimal basis**?

1. Identify which variable represents **ponies**: ponies (x2)
2. In the final table find column related to ponies:

1,5
0
-0,5

3. Copy **final RHS**:

0,7
0,2
0,1

4. Divide final RHS by column of associated change (column related to ponies in final table):

$0,7 : 1,5 = 0,46666$ (Choose the lowest positive just like in the test for feasibility)

$0,2 : 0 = \text{none}$

$0,1 : (-0,5) = -0,2$

d) How will **change the solution** if the **supply of material** goes down by 10 kg? and if goes up?

1. Identify which variable represents **supply of material**: supply of material (x4)
2. In the final table find column related to **supply of material**:

0,005
0
-0,005
0,03

3. Copy **final RHS**:

0,7
0,2
0,1

4. Change

- Less material -> less toys -> smaller profit -> -10
- More material -> more toys -> higher profit -> +10

5. Proceed calculation:
a)

0,7		0,005		0,65
0,2	+10*	0	=	0,2
0,1		-0,005		0,15
5		0,03		4,7

- b)

0,7		0,005		0,75
0,2	-10*	0	=	0,2
0,1		-0,005		0,05
5		0,03		5,3

e) Calculate the maximal change in the **availability of material** (**without changing the basis**)!

1. Identify which variable represents **supply of material**: supply of material (x4)
2. In the final table find column related to **supply of material**:

0,005
0
-0,005

3. Copy **final RHS**:

0,7
0,2
0,1

4. Divide final RHS by column of associated change (column related to the **availability of material** in final table):

$0,7 : 0,005 = 140$ (b1 can go down to $240 - 140 = 100$)

$0,2 : 0 = \text{forbidden}$

$0,1 : -0,005 = 20$ (b1 can go up to $240 + 20 = 260$)

5. Proceed calculation:

a)

0,7		0,005		0
0,2	+140*	0	=	0,2
0,1		-0,005		0,8
5		0,03		0,8

b)

0,7		0,005		0,8
0,2	-20*	0	=	0,2
0,1		-0,005		0
5		0,03		5,6

f) What must be the minimal price of the ponies so that they worth producing?

1. Identify which variable represents ponies: ponies (x_2)
2. In the final table find column related to ponies:

1,5
0
-0,5
2 – shadow price

3. Identify original price of ponies (x_2) = 7 (c_2 from initial table)
4. Sum up original price and shadow price = 9 (If the price reaches the exat 9 ($7+2$), there will be alternative solution), so 9+ per piece are worth producing.

[illegible]

- b) Yellow cell represents profit (fill formula to the cell)

SUMA ✖ ✓ f_x =C11*C12+D11*D12+E11*E12									
	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8									
9	This is how to prepare the model								
10									
11		Prices	6	7	4				
12									
13			elephants	girafes	ponies	LHS	RHS		
14		wood	200	300	500	0	260		
15		giraffes	0	0	1	0	0,2		
16		total cap.	1	1	1	0	1		
17									
18						Profit	+E11*E12		
19									
20									
21									

- Click on Solver add-in under Data tab
- Fill in the form of Solver Parameter, then click on Solve

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

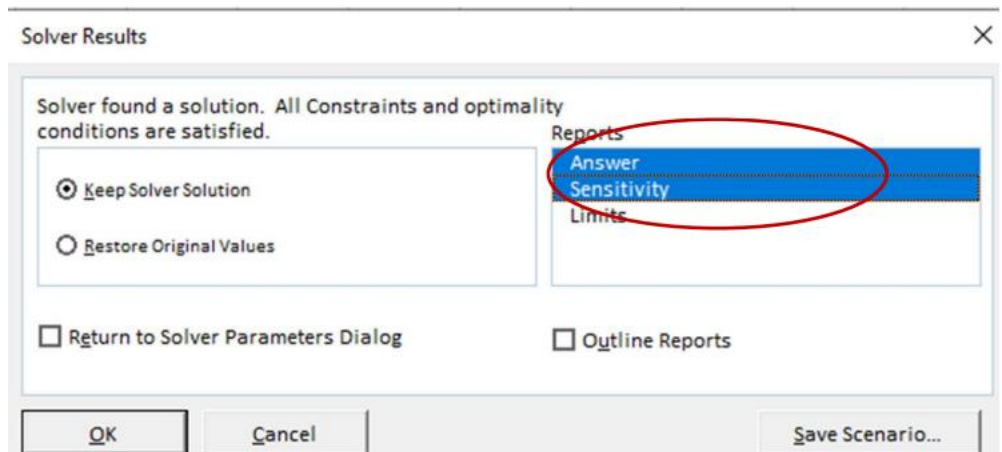
\$F\$14 <= \$G\$14
 \$F\$15 >= \$G\$15
 \$F\$16 <= \$G\$16

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method: **Simplex LP**

Solving Method
 Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

5. Choose Answer and Sensitivity reports



1. Interpret the solution from the solver output

a) Answer report

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

Solver Engine

Engine: Simplex LP
 Solution Time: 0,031 Seconds.
 Iterations: 4 Subproblems: 0

Solver Options

Max Time Unlimited, Iterations Unlimited, Precision 0,000001, Use Automatic Scaling
 Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Max)

Cell	Name	Original Value	Final Value
\$G\$18	Profit RHS	0	5

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$C\$12		0	0,7	Contin
\$D\$12		0	0	Contin
\$E\$12		0	0,2	Contin

these are the right hand sides of the final solution

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$F\$14	wood LHS	240	\$F\$14<=\$G\$14	Binding	0
\$F\$15	giraffes LHS	0,2	\$F\$15>=\$G\$15	Binding	0
\$F\$16	total cap. LHS	0,9	\$F\$16<=\$G\$16	Not Binding	0,1

x6 in the hand solution slack variable showing that not the whole capacity is used

b) Sensitivity report

Microsoft Excel 16.0 Sensitivity Report
 Worksheet: [tutorial1_solution.xlsx]task2
 Report Created: 27.01.2023 14:08:11

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$12		0,7	0	6	1E+30	1,333333333
\$D\$12		0	-2	7	2	1E+30
\$E\$12		0,2	0	4	11	1E+30

If we change the objective function coefficients in this interval, the basis will not change (but the value of the objective function will change if we change the coefficient of the basic variables!)

this is infinity --- no bound, if we decrease the price of ponies, basis will not change (they are not in the basis, its not a surprise then)

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$F\$14	wood LHS	240	0,03	240	20	140
\$F\$15	giraffes LHS	0,2	-11	0,2	0,28	0,066666667
\$F\$16	total cap. LHS	0,9	0	1	1E+30	0,1

Same, but for right hand sides

2. Increase the amount of the wood that it will change the basis (what is the minimum?) and solve with the Solver again.

1. rewrite the first RHS from 240 to 261

This is how to prepare the model

	Prices	6	7	4		
		elephants	girafes	ponies	LHS	RHS
	wood	200	300	500	0	261
	giraffes	0	0	1	0	0,2
	total cap.	1	1	1	0	1
					Profit	0

2. Proceed the same steps in Solver

3. Interpret the results and answer the questions:

In case **availability of material** increases to 261, will it change the basis?

3. Change the amount of the wood back to 240. Change the price of the elephants to 4 and solve again

1. Change the RHS1 back to 240

2. Change the price of elephants from 6 to 4 and check the result

	Prices	4	7	4		
		elephants	girafes	ponies	LHS	RHS
	wood	200	300	500	0	240
	giraffes	0	0	1	0	0,2
	total cap.	1	1	1	0	1
					Profit	0

2. Proceed the same steps in Solver

3. Interpret the results and answer the question:

In case **price of elephant** increases to 6, will it change the basis?